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A simple technique to measure the magnetic susceptibility of liquids

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A novel technique for measuring the magnetic susceptibility of liquids with modest applied magnetic fields (order of 0.25 T) is presented. The deformation of the liquid surface by a magnetic field is determined by a laser bounce or optical lever technique. The energy balance between the magnetic energy and the gravitational potential of the diamagnetic or paramagnetic liquid interaction is used to determine the susceptibility. The energy due to the surface tension energy is about 10% of the gravitational energy and can be neglected for fast measurements. © *2012 American Institute of Physics*. [http://dx.doi.org/10.1063/1.4749847]

I. INTRODUCTION

The magnetic properties of liquids are most often determined by magnetic measurements on bulk liquids using rather expensive equipment such as vibrating sample or SQUID magnetometers. Though not usually quantitative there are also techniques to illustrate visually the magnetic properties of liquids with water the most common liquid for illustration. Perhaps the most famous demonstration of the diamagnetism in water is the levitation of a frog requiring fields on the order of 16 T.[1] Ueno's group performed more quantitative experiments by deforming a liquid surface using applied fields on the order of 4 T and dubbed the phenomenon the "Moses effect."[2,3] Both of these experiments require rather specialized equipment, including superconducting magnets, rendering them unfeasible in many situations. More accessible experiments have also been performed in the past, though these have been qualitative in nature.[4–6]

We previously demonstrated a technique similar to Ueno's experiment but required only the modest magnetic fields produced by NdFeB permanents magnets using an optical lever technique.[7] In the current work, we expand on our previous effort to relate quantitatively the magnetic, gravitational and surface tension energies to provide a new method for determining the magnetic susceptibility of liquids. Examples of the utility of the technique include measurements of water, saline solutions, and paramagnetic copper sulfate. This novel technique provides several benefits for a determination of the magnetic susceptibility in that the basic apparatus is cheap (costs on the order of $200), is relatively fast (takes about 1/2 h to determine the susceptibility), and is accurate to about 10% in its easiest implementation.

II. THEORY

Using water as an example, the diamagnetic response of water in the presence of a magnetic field forces the surface of the water away from the magnetic field. In the case of water in a dish, such as shown in our experimental setup in Figure 1, the surface would curve downward, forming a depression on the surface regardless of the direction of the field; the water is repelled by the magnetic field. The surface shape is determined by the requirement the energy is constant on the surface. In a liquid, this energy surface is a minimum of the gravitational, magnetic, and surface tension energies. We have found the gravitational potential and magnetic energy to be dominant for the liquids we investigated. For changes in the surface such that the magnetic field is a constant the gravitational energy is given by

$$\Delta u_{grav} = \rho g \Delta h \tag{1}$$

and the magnetic energy is given by

$$\Delta u_{mag} = -\int_0^H \mu_0 \chi_v H^* dH^* = -\frac{\mu_0 \chi_v H^2}{2}, \tag{2}$$

where Δh is the displacement of the surface, ρ is mass density of the liquid, μ_0 is the permeability of free space, g is the gravitational acceleration, and χ_v is the volume magnetic susceptibility. Since the energy of these two must offset each other,

$$\Delta u_{total} = \Delta u_{grav} + \Delta u_{mag} = 0. \tag{3}$$

We can use Eqs. (1) and (2) in Eq. (3) to find

$$\Delta h = \frac{\mu_0 H^2}{2g} \frac{\chi_v}{\rho}, \tag{4}$$

where we are assuming a uniform applied magnetic field strength over the displaced volume as stated earlier. As is clear from Eq. (4) the deflection is proportional to the susceptibility divided by the density. For water and other diamagnetic materials χ_v is negative, so in order for u_{total} to be constant, the contribution from u_{grav} must be negative, and so the water surface must form a depression. For a paramagnetic solution, the magnetic energy density term is negative, so the solution will form a bulge in the surface.

III. APPARATUS AND MEASUREMENT TECHNIQUE

A petri dish with the liquid sample filled to a depth of approximately 0.5 cm was set on top of a frame constructed from plastic though in principle any non-ferromagnetic material could be used. A magnetic field on the order of 2 kOe was generated by a PASCO EM-8641 Variable gap magnet, which consisted of two NdFeB magnets each 1 cm in diameter, and

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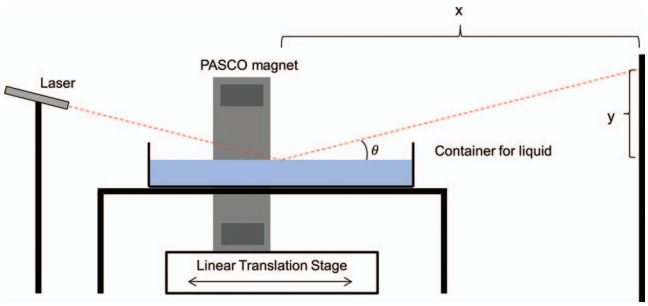

FIG. 1. Cartoon of experimental setup. The PASCO magnet has a U-shaped yoke with two NdFeB magnets attached at the open end of the U. The dark rectangles in the picture represent the NdFeB magnets and the region between them is the air gap. Note that the figure is not to scale.

set approximately 6 cm apart. Such a configuration was chosen to minimize the variation in field strength as a function of height; a single magnet under the petri dish alters the surface but the stronger field variation with distance from the magnet increases the field error. A cartoon of the apparatus as described is shown in Figure 1. The profile of the deformation of a liquid sample was measured using a 5 mW 670 nm laser, which on an undistorted surface of a liquid sample had an incident angle, θ, on the order of 15°. When the water was deformed by the magnetic field, the incident angle of the laser beam was shifted by an angle $\Delta\theta$, which resulted in a shift of the reflection angle by $2\Delta\theta$. The entire setup was placed 8 m from a wall, such that a shift in the reflected angle of the laser beam would show up as a corresponding shift in the height Δy of where the laser beam hits the wall, which was measured using a ruler. Simple trigonometry gives the follow relation for the change in reflection angle:

$$2\Delta\theta = \tan^{-1}\left(\frac{y+\Delta y}{x}\right) - \theta. \tag{5}$$

By measuring this quantity as a function of position, we obtained the slope of the sample surface at every point, giving us a profile of the surface. To keep y and x constant in Eq. (5) we shifted the depression itself relative to the incident beam by setting the PASCO magnet on a translating stage. The magnet was shifted along the translation axis in increments of 1 mm. The corresponding magnetic field at each point was measured using a gaussmeter set at the same position relative to the magnets as the water surface. The surface profile measurement was repeated for 5 different solutions, listed in Table I along with their relevant physical properties.

TABLE I. Solutions used in the experiment and their relevant physical properties.

Solution	Density (g/cm^3)	Magnetic susceptibility[a]
Water	0.997	-9.05×10^{-6}
Saturated NaCl solution	1.32	-1.11×10^{-5}
Copper sulfate, 25% saturation	1.09	-3.17×10^{-6}
Copper sulfate, 50% saturation	1.18	2.71×10^{-6}
Copper sulfate, 100% saturation	1.36	1.45×10^{-5}

[a]Reference 8.

IV. RESULTS AND DISCUSSION

Figure 2(a) shows the result of a measurement of the reflection angle of pure water under the influence of a magnetic field; the corresponding magnetic field is shown in Figure 2(b). For our arrangement the maximum magnetic field was 2.35 kOe.

In order to compare the surface profile with the measured magnetic fields, we see from Eq. (2) that we must obtain the displacement of the water as a function of position. In the small angle approximation, the change in angle of the water surface can be taken to be the slope of the water, and thus the displacement can be obtained by performing a Riemann sum on the measured data. For our calculation, we used the trapezoidal rule for Riemann summing, and summed both left to right and right to left and took the average. Equation (2) predicts that the displacement is proportional to the square of the applied magnetic field. The result of the summation for several different solutions is shown in Figure 3(a), where the square of the applied magnetic field profile that was used for all of the solutions is shown in Figure 3(b). Note the y-axis in Figure 3(a) is in microns compared to the x-axis in mm and thus the surface displacement is greatly enhanced.

To test the accuracy of the technique we compared the data taken with multiple solutions to the same magnetic field profile using Eq. (4) which gives a linear relationship between the quantities χ_v/ρ and Δh for each field value. Figure 4 is a plot of the measured values versus the known values for χ_v/ρ for each solution against the maximum change in height of the deformed solution.

Uncertainties were estimated by measuring deflection vs. position multiple times, and were found to be approximately 10%. A linear fit shows the slope of the plot to be 1945.0 ± 65.0 kg m^2, which corresponds to the value $\frac{\mu_0 H^2}{2g}$, which has a calculated value of 2240 kg m^2. For this set of data, the measured slope is about 13% low which is most likely due to some combination of magnetic field error and neglecting the surface tension energy. If all the error were due to the value of magnetic field it would mean translate into a 6% error in the field values. Taking the opposite approach, we will next consider the surface tension energies.

To determine the surface tension energies we measured the contact angle of each solution on a teflon surface using a Kyowa MCA-3 Image Analysis Contact Angle Meter. A comparison of the relevant parameters, including the surface tension, is given in Table II which includes the value for the corrected Δh with the surface tension included. In this table to include the effect of surface tension, we calculated the total surface tension energy for each profile by numerically integrating the surface area of the surface of revolution of the profile and subtracting off the area of a flat circle filling the same space. For comparison, we determined the total change in gravitational energy for each profile by numerically computing the volume integral of the gravitational energy density. Because surface tension always resists the deformation of a surface, the sum in absolute value of the volume energy and the surface energy is equivalent to the total energy available to form the surface. We estimated the effect of surface tension on the Δh values of a given profile by scaling the values by

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TABLE II. A comparison of the relevant parameters, including the surface tension, is given for each of the solutions used in the experiment.

Solution	Surface tension (N/m)	$\Delta E_{surface}$ (J)	ΔE_{grav} (J)	$\Delta E_{surface}/\Delta E_{grav}$	Rescaling factor for Δh
Water	0.073	4.17×10^{-11}	5.51×10^{-10}	7.03%	3.7%
NaCl, 100% saturation	0.113	5.50×10^{-11}	5.97×10^{-10}	9.20%	4.5%
$CuSO_4$, 25% saturation	0.079	4.98×10^{-12}	6.12×10^{-11}	8.13%	4.0%
$CuSO_4$, 50% saturation	0.078	2.24×10^{-12}	4.14×10^{-11}	5.40%	2.7%
$CuSO_4$, 100% saturation	0.097	8.34×10^{-11}	9.44×10^{-10}	8.83%	4.5%

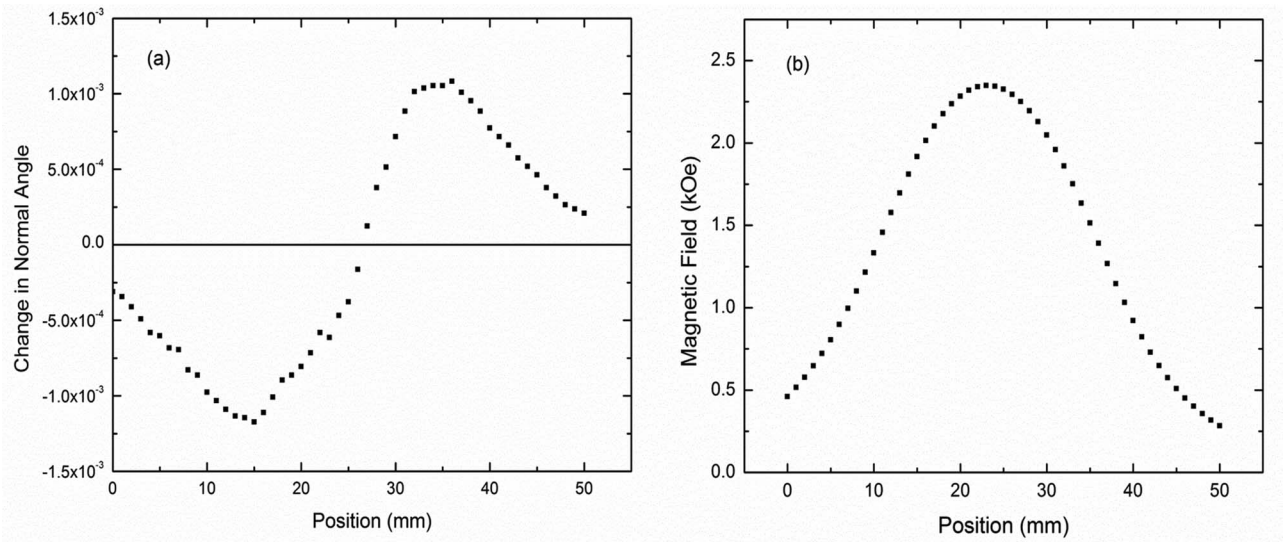

FIG. 2. Panel (a) shows the slope of the liquid's surface as a function of the position of the magnet. The zero value for the magnet's position is arbitrary and for the data shown corresponds to the magnet edge approximately directly below the laser spot on the liquid's surface; 50 mm corresponds to the other edge of the magnet. Panel (b) plots the value of the applied magnetic field at approximately the liquid's surface.

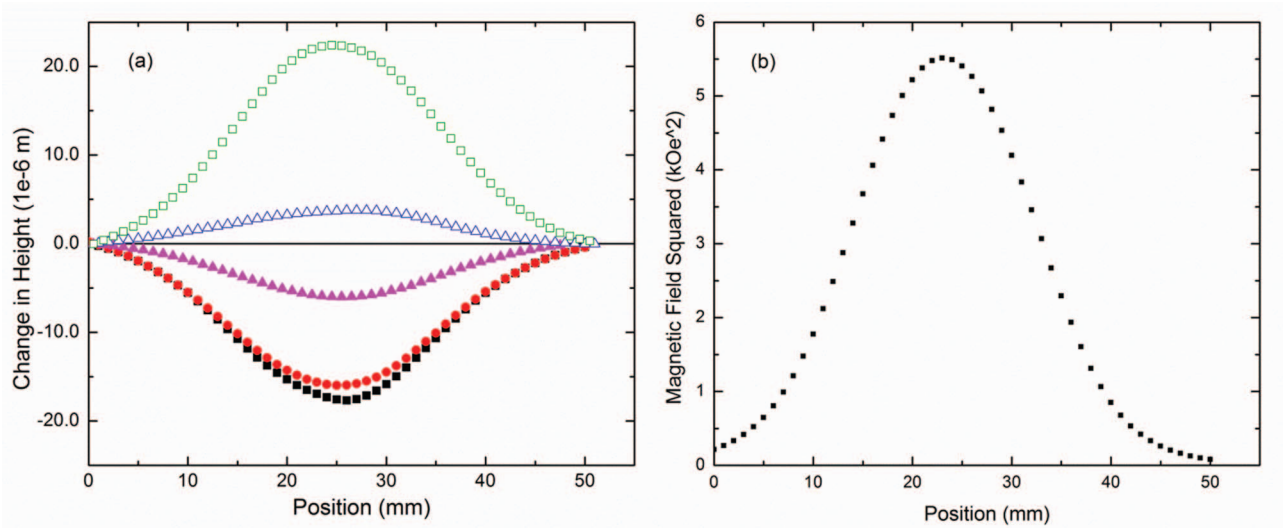

FIG. 3. (a) Shown here is the vertical displacement of several liquids' surface as a function of the position of the magnet. In this figure the individual curves correspond to water (filled black squares), 100% NaCl solution (filled red circles), 25% CuSO4 solution (filled magenta triangles), 50% CuSO4 solution (unfilled blue triangles), and 100% CuSO4 solution (unfilled green squares). The vertical displacements are a function of the square of the magnetic field which is shown in (b).

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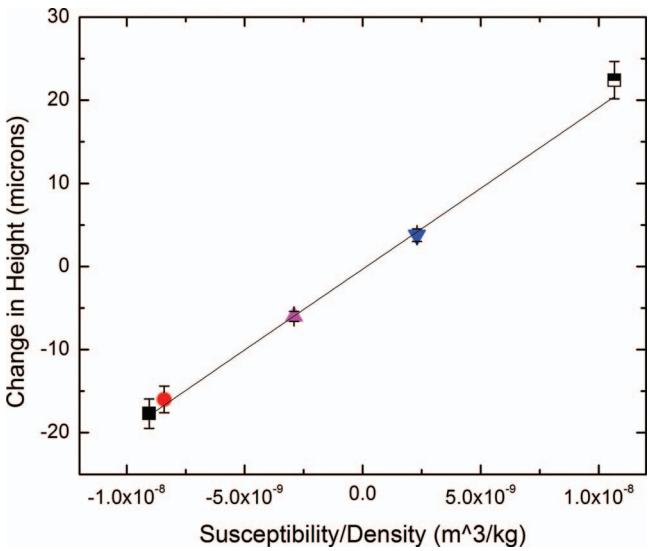

FIG. 4. Plot of the liquid surface maximum displacement versus the ratio of the magnetic susceptibility to the mass density of the liquids. Each data point corresponds to the maximum displacement for one of the following liquid surface profiles: water (filled black square), 100% NaCl solution (filled red circle), 25% CuSO4 solution (filled magenta triangle), 50% CuSO4 solution (filled inverted blue triangle), and 100% CuSO4 solution (half filled black square).

a constant multiplicative factor and recalculating the volume integral for the gravitational energy. The scaling factor was then adjusted until the total gravitational energy equaled the total energy in the profile in the previous calculation. In other words, we scaled each profile to estimate a hypothetical profile where surface tension is zero and does not contribute to the energy. A table showing the results of our calculations is shown in Table II.

We observe that the surface tension has approximately a 5% effect on the heights of the surface profile. While the correction is in the right direction, as it increases the absolute value of all Δh_{max}, it does not completely account for the discrepancy in the slope of Δh_{max} versus χ_v/ρ. Taking into account the effect of the surface tension would reduce the error of the magnetic field to about 3% or 4%.

V. SUMMARY

In summarizing our work, we have presented a relatively simple technique based on an optical lever to determine the magnetic susceptibility of liquids; as has happened many times in the history of science, an optical lever can magnify a rather small displacement and the present technique is yet another example. In practice the surface can be accurately measured in about half an hour and the analysis takes only a few minutes more. The total cost of the system makes it very inexpensive compared to other magnetometery systems. These comments apply if we are only including magnetic and gravitational energies of the system and neglecting the surface tension. This approach for the liquid examples we used gave us an accuracy on the order of 10% which includes the errors in the magnetic field measurement and from neglecting the surface tension. As shown, the maximum change in height of a deformation in a given solution was found to be linearly related to the quantity χ_v/ρ, which also followed from the model.

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